

5.1. THEORY SAYS EQUITY RISK PREMIUMS ARE PREDICTABLE

Consider the Gordon (1963) *Dividend Discount Model*, which states that the equity price, P , is the present value of future discounted dividends:

$$P = \frac{D}{E(r) - g}, \quad (8.2)$$

where D is the expected dividend next period, $E(r)$ is the discount rate, and g is the growth rate of dividends. Equation (8.2) is a valuation formula. Holding D fixed, prices today are high because $1/(E(r) - g)$ is low: either future expected returns are low, future growth is high, or both. The common wisdom is that high prices occur because growth will be high. For example, at the time of writing, the high share prices of Apple Inc. (AAPL) and Facebook Inc. (FB) suggest that the future growth of these companies will continue. But generally, high prices forecast low growth. (This is the *value effect*; see chapter 7).

The Gordon model in equation (8.2) can be equivalently expressed as

$$\frac{D}{P} = E(r) - g, \quad (8.3)$$

which states that high dividend yields are caused by high future expected returns, low future growth, or both. We could also write

$$E(r) = \frac{D}{P} + g, \quad (8.4)$$

which states that the expected return is the dividend yield plus the cash flow growth. Equation (8.4) says that dividend yields should help forecast expected returns, and this has been examined in finance since Dow (1920).

Academics have examined *variance decompositions* of dividend yields, which are computed by taking variances of both sides of equation (8.3). By doing so we can decompose movements of dividend yields into movements due to expected returns or discount rates, $E(r)$, or movements due to g , and movements due to the co-movement of both:

$$\text{var}\left(\frac{D}{P}\right) = \text{var}(E(r)) + \text{var}(g) - 2\text{cov}(E(r), g).$$

For the discussion below, let's assume that you can ignore the covariance term so we write:¹⁸

$$\text{var}\left(\frac{D}{P}\right) \approx \text{var}(E(r)) + \text{var}(g). \quad (8.5)$$

¹⁸ Serious studies do not ignore the covariance term. They assign the covariance term to the other variance terms through orthogonalization (usually Cholesky) procedures.